

Sub phase -2 Technical stack

Date	28-09-2026
Team ID	SWTID-2026-7624
Project name	Script-controlled ACL
Maximum marks	4 marks

4. Technical Stack

The following technologies and components are used for implementing the Script-Controlled ACL project.

Component| Technology / Tool

Platform| ServiceNow

Access Control| Script-Controlled ACL

Programming Language| JavaScript

Database| ServiceNow Database

User Management| ServiceNow User & Role Management

Security| ACL / Role-Based Access Control

Testing| ServiceNow Impersonation

Data Table| Institution Details

Table Name| u_institution_details

Database Structure

The project uses a custom Institution Details table.

Table Label:

Institution Details

Table Name:

u_institution_details

Fields:

- Student Roll Number – Auto Number
- Student Name – Reference
- Faculty Name – Reference
- Branch – Choice (ECE, EEE, CSE)
- Email – String
- Phone Number – String
- Description – Multi-Line String

The Phase-1 design specifies these fields and the custom roles bb1, bb2, bb3, and bb4 for different CRUD permissions.

4. Technical Stack

The following technologies and components are used for implementing the Script-Controlled ACL project.

Component	Technology / Tool
Platform	ServiceNow
Access Control	Script-Controlled ACL
Programming Language	JavaScript
Database	ServiceNow Database
User Management	ServiceNow User & Role Management
Security	ACL / Role-Based Access Control
Testing	ServiceNow Impersonation
Data Table	Institution Details
Table Name	u_institution_details

Database Structure

Table Label : Institution Details	Fields																								
Table Name : u_institution_details	<table><tr><th>No.</th><th>Field Name</th><th>Type</th></tr><tr><td>1</td><td>Student Roll Number</td><td>Auto Number</td></tr><tr><td>2</td><td>Student Name</td><td>Reference (sys_user)</td></tr><tr><td>3</td><td>Faculty Name</td><td>Reference (sys_user)</td></tr><tr><td>4</td><td>Branch</td><td>Choice (ECE, EEE, CSE)</td></tr><tr><td>5</td><td>Email</td><td>String</td></tr><tr><td>6</td><td>Phone Number</td><td>String</td></tr><tr><td>7</td><td>Description</td><td>Multi-Line String</td></tr></table>	No.	Field Name	Type	1	Student Roll Number	Auto Number	2	Student Name	Reference (sys_user)	3	Faculty Name	Reference (sys_user)	4	Branch	Choice (ECE, EEE, CSE)	5	Email	String	6	Phone Number	String	7	Description	Multi-Line String
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Conclusion

Phase-2 Requirement Analysis defines how the Script-Controlled ACL system will operate from the user's perspective and identifies the functional, non-functional, data-flow, and technical requirements.

The proposed solution provides controlled access to records by combining user roles with field-based access conditions. This approach provides more flexible and granular security than relying only on